

The Binomial Theorem

Name
Date
Course

Exponent Rules

I) $x^a \cdot x^b = x^{a+b}$ Example: $x^2 \cdot x^5 = x^{2+5} = x^7$

II) $\frac{x^a}{x^b} = x^{a-b}$, $x \neq 0$ Example: $\frac{x^9}{x^3} = x^{9-3} = x^6$

III) $(xy)^a = x^a \cdot y^a$ Example: $(2m)^4 = 2^4 \cdot m^4$

IV) $\left(\frac{x}{y}\right)^a = \frac{x^a}{y^a}$, $y \neq 0$ Example: $\left(\frac{n}{3}\right)^7 = \frac{n^7}{3^7}$

V) $(x^a)^b = x^{ab}$ Example: $(x^{10})^3 = x^{10 \cdot 3} = x^{30}$

VI) $x^1 = x$ Example: $3^1 = 3$

VII) $x^0 = 1$, $x \neq 0$ Example: $5^0 = 1$

VIII) $x^{-a} = \frac{1}{x^a}$, $x \neq 0$ Example: $x^{-2} = \frac{1}{x^2}$

IX) $x^{\frac{1}{a}} = \sqrt[a]{x}$ Example: $x^{\frac{1}{3}} = \sqrt[3]{x}$

X) $x^{\frac{b}{a}} = (\sqrt[a]{x})^b$ or $\sqrt[a]{x^b}$ Example: $x^{\frac{5}{7}} = (\sqrt[7]{x})^5$ or $\sqrt[7]{x^5}$

Review on Exponent Rule III.

Simplify.

• Ex 1: $(5x)^3 = 5^3 \cdot x^3 = \boxed{125x^3}$

Ex 2: $(-6y)^2 = (-6)^2 y^2 = \boxed{36y^2}$

• Ex 3: $(2a)^5 = 2^5 \cdot a^5 = \boxed{32a^5}$

Ex 4: $(-3b)^3 = (-3)^3 b^3 = \boxed{-27b^3}$

Review on Exponent Rule IV.

Simplify.

• Ex 5: $(x^3)^2 = x^{3 \cdot 2} = \boxed{x^6}$

Ex 6: $(a^4)^6 = \boxed{a^{24}}$

• Ex 7: $(y^{10})^2 = \boxed{y^{20}}$

Ex 8: $(b^5)^6 = \boxed{b^{30}}$

$$(xy)^a = x^a \cdot y^a \quad (x^a)^b = x^{ab}$$

Using Exponent Rules III & IV

Simplify.

• Ex 9: $(2x^6)^3 = 2^3 \cdot (x^6)^3 = \boxed{8x^{18}}$

Ex 10: $(-4a^2)^2 = (-4)^2 (a^2)^2 = \boxed{16a^4}$

• Ex 11: $(3y^4)^4 = 3^4 (y^4)^4 = \boxed{81y^{16}}$

Ex 12: $(-10b^3)^2 = (-10)^2 (b^3)^2 = \boxed{100b^6}$

Alternate Notation for nCr

$$nCr = C(n, r) = \binom{n}{r} = \frac{n!}{(n-r)!r!}$$

Practice

Evaluate.

• Ex 13: $\binom{3}{1} = {}_3C_1 = \boxed{3}$

Ex 14: $\binom{5}{2} = {}_5C_2 = \boxed{10}$

• Ex 15: $\binom{7}{4} = {}_7C_4 = \boxed{35}$

Ex 16: $\binom{6}{5} = \boxed{6}$

Zero Factorial Is Defined as 1: $0! = 1$

• Ex 17: $\binom{4}{0} = {}_4C_0 = \boxed{1}$

Ex 18: $\binom{2}{0} = \boxed{1}$

• Ex 19: $\binom{9}{0} = \boxed{1}$

Ex 20: $\binom{8}{0} = \boxed{1}$

The Binomial Theorem

$$(a+b)^n = ?$$

$$(a+b)^0 = \boxed{1}$$

$$(a+b)^1 = \boxed{a+b}$$

$$(a+b)^2 = (a+b)(a+b) = a^2 + ab + ba + b^2 = \boxed{a^2 + 2ab + b^2}$$

$$(a+b)^3 = (a+b)(a+b)(a+b) = (a^2 + 2ab + b^2)(a+b)$$

$$= a^3 + a^2b + 2a^2b + 2ab^2 + ab^2 + b^3$$

$$= \boxed{a^3 + 3a^2b + 3ab^2 + b^3}$$

The Expansion Progression

$$\begin{aligned}
 (a+b)^0 &= 1 \\
 (a+b)^1 &= a + b \\
 (a+b)^2 &= (a+b)(a+b) = a^2 + 2ab + b^2 \\
 (a+b)^3 &= a^3 + 3a^2b + 3ab^2 + b^3 \\
 (a+b)^4 &= a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + b^4 \\
 (a+b)^5 &= a^5 + 5a^4b + 10a^3b^2 + 10a^2b^3 + 5ab^4 + b^5
 \end{aligned}$$

Pascal's Triangle

$$\begin{array}{ccccccc}
 & & & & 1 & & & & \\
 & & & 1 & & 1 & & & \\
 & & 1 & & 2 & & 1 & & \\
 & 1 & & 3 & & 3 & & 1 & \\
 1 & & 4 & & 6 & & 4 & & 1 \\
 & 1 & & 5 & & 10 & & 10 & & 5 & & 1 \\
 & & 1 & & 6 & & 15 & & 20 & & 15 & & 6 & & 1 \\
 1 & & & 7 & & 21 & & 35 & & 35 & & 21 & & 7 & & 1
 \end{array}$$

Determining Coefficients with Counting Algorithms Learned in Previous Classes

$$(a+b)^4 = (a+b)(a+b)(a+b)(a+b)$$

$$\begin{aligned}
 &= aaaa + aaab + aaba + abaa + ba aa + aabb + abba \\
 &+ bbaa + baab + baba + abab + abbb + babb + bbab + bbba + bbbb \\
 &= 1a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + 1b^4
 \end{aligned}$$

$$= {}_4C_0 a^4 + {}_4C_1 a^3b + {}_4C_2 a^2b^2 + {}_4C_3 ab^3 + {}_4C_4 b^4$$

$$(a+b)^n = {}_nC_0 a^n b^0 + {}_nC_1 a^{n-1} b^1 + {}_nC_2 a^{n-2} b^2 + \dots + {}_nC_n a^0 b^n$$

Note: To avoid messy subscripts in this expression, ${}_nC_r$ is often written as $\binom{n}{r}$.

So we have learned 3 ways to write the number of groups of "r" distinct things that can be formed from "n" distinct things if the order (i.e. permutations) of the "r" things does not matter.

$${}_nC_r = C(n, r) = \binom{n}{r} = \frac{n!}{(n-r)!r!}$$

The Binomial Theorem

$$(a+b)^n = \binom{n}{0} a^n b^0 + \binom{n}{1} a^{n-1} b^1 + \binom{n}{2} a^{n-2} b^2 + \dots + \binom{n}{n} a^0 b^n$$

"a" and "b" must be real numbers and n must be a positive integer.

Practice

Use the Binomial Theorem to expand the following expressions.

① $(x+2)^6$

$$\binom{6}{0}x^6(2^0) + \binom{6}{1}x^5(2^1) + \binom{6}{2}x^4(2^2) + \binom{6}{3}x^3(2^3) + \binom{6}{4}x^2(2^4) + \binom{6}{5}x(2^5) + \binom{6}{6}x^0(2^6)$$

$$(1)x^6(1) + 6x^5(2) + 15x^4(4) + 20x^3(8) + 15x^2(16) + 6x(32) + (1)(1)(64)$$

$$x^6 + 12x^5 + 60x^4 + 160x^3 + 240x^2 + 192x + 64$$

② $(y-3)^5 = (y+(-3))^5$

$$\binom{5}{0}y^5(-3)^0 + \binom{5}{1}y^4(-3)^1 + \binom{5}{2}y^3(-3)^2 + \binom{5}{3}y^2(-3)^3 + \binom{5}{4}y(-3)^4 + \binom{5}{5}y^0(-3)^5$$

$$(1)y^5(1) + 5y^4(-3) + 10y^3(9) + 10y^2(-27) + 5y(81) + (1)(1)(-243)$$

$$y^5 - 15y^4 + 90y^3 - 270y^2 + 405y - 243$$

③ $(k+4)^3$

$$\binom{3}{0}k^3(4^0) + \binom{3}{1}k^2(4^1) + \binom{3}{2}k(4^2) + \binom{3}{3}k^0(4^3)$$

$$(1)k^3(1) + (3)k^2(4) + 3k(16) + (1)(1)(64)$$

$$k^3 + 12k^2 + 48k + 64$$

④ $(m+6)^5$

$$\binom{5}{0}m^5(6^0) + \binom{5}{1}m^4(6^1) + \binom{5}{2}m^3(6^2) + \binom{5}{3}m^2(6^3) + \binom{5}{4}m(6^4) + \binom{5}{5}m^0(6^5)$$

$$(1)m^5(1) + 5m^4(6) + 10m^3(36) + 10m^2(216) + 5m(1296) + (1)(1)(7,776)$$

$$m^5 + 30m^4 + 360m^3 + 2,160m^2 + 6,480m + 7,776$$

⑤ $(x+1)^4$

$$\binom{4}{0}x^4(1^0) + \binom{4}{1}x^3(1^1) + \binom{4}{2}x^2(1^2) + \binom{4}{3}x(1^3) + \binom{4}{4}x^0(1^4)$$

$$(1)x^4(1) + 4x^3(1) + 6x^2(1) + 4x(1) + (1)(1)(1)$$

$$x^4 + 4x^3 + 6x^2 + 4x + 1$$

⑥ $(t-4)^6 = (t+(-4))^6$

$$\binom{6}{0}t^6(-4)^0 + \binom{6}{1}t^5(-4)^1 + \binom{6}{2}t^4(-4)^2 + \binom{6}{3}t^3(-4)^3 + \binom{6}{4}t^2(-4)^4 + \binom{6}{5}t(-4)^5 + \binom{6}{6}t^0(-4)^6$$

$$(1)t^6(1) + 6t^5(-4) + 15t^4(16) + 20t^3(-64) + 15t^2(256) + 6t(-1024) + (1)(1)(4096)$$

$$t^6 - 24t^5 + 240t^4 - 1280t^3 + 3840t^2 - 6144t + 4096$$

• ⑦ $(2m-5p)^4 = (2m+(-5p))^4$

$$\binom{4}{0}(2m)^4(-5p)^0 + \binom{4}{1}(2m)^3(-5p)^1 + \binom{4}{2}(2m)^2(-5p)^2 + \binom{4}{3}(2m)^1(-5p)^3 + \binom{4}{4}(2m)^0(-5p)^4$$

$$(1)(16m^4)(1) + 4(8m^3)(-5p) + 6(4m^2)(25p^2) + 4(2m)(-125p^3) + (1)(1)(625p^4)$$

$$\boxed{16m^4 - 160m^3p + 600m^2p^2 - 1000mp^3 + 625p^4}$$

• ⑧ $(3K+2L)^3$

$$\binom{3}{0}(3K)^3(2L)^0 + \binom{3}{1}(3K)^2(2L)^1 + \binom{3}{2}(3K)^1(2L)^2 + \binom{3}{3}(3K)^0(2L)^3$$

$$(1)27K^3(1) + 3(9K^2)(2L) + 3(3K)(4L^2) + (1)(1)(8L^3)$$

$$\boxed{27K^3 + 54K^2L + 36KL^2 + 8L^3}$$

• ⑨ $(x^2+3y^4)^3$

$$\binom{3}{0}(x^2)^3(3y^4)^0 + \binom{3}{1}(x^2)^2(3y^4)^1 + \binom{3}{2}(x^2)^1(3y^4)^2 + \binom{3}{3}(x^2)^0(3y^4)^3$$

$$(1)x^6(1) + 3x^4(3y^4) + 3x^2(9y^8) + (1)(1)(27y^{12})$$

$$\boxed{x^6 + 9x^4y^4 + 27x^2y^8 + 27y^{12}}$$

• ⑩ $(2a^3-b^6)^5 = (2a^3+(-b^6))^5$

$$\binom{5}{0}(2a^3)^5(-b^6)^0 + \binom{5}{1}(2a^3)^4(-b^6)^1 + \binom{5}{2}(2a^3)^3(-b^6)^2 + \binom{5}{3}(2a^3)^2(-b^6)^3 + \binom{5}{4}(2a^3)^1(-b^6)^4 + \binom{5}{5}(2a^3)^0(-b^6)^5$$

$$(1)(32a^{15})(1) + 5(16a^{12})(-b^6) + 10(8a^9)(b^{12}) + 10(4a^6)(-b^{18}) + 5(2a^3)(b^{24}) + (1)(1)(-b^{30})$$

$$\boxed{32a^{15} - 80a^{12}b^6 + 80a^9b^{12} - 40a^6b^{18} + 10a^3b^{24} - b^{30}}$$

• ⑪ Find the coefficient of the x^5 term in the expansion of $(x+3)^7$

$$\binom{7}{0}x^7(3^0) + \binom{7}{1}x^6(3^1) + \binom{7}{2}x^5(3^2) + \dots + \binom{7}{7}x^0(3^7)$$

$$\downarrow$$

$$21x^5(9)$$

$$\downarrow$$

$$189x^5$$

$$\downarrow$$

$$\boxed{189}$$

⑫ Find the coefficient of the k^{25} term in the expansion of $(y-k^5)^8$.

$$(y-k^5)^8 = (y+(-k^5))^8$$

$$\binom{8}{0}y^8(-k^5)^0 + \dots + \binom{8}{5}y^3(-k^5)^5 + \binom{8}{6}y^2(-k^5)^6 + \binom{8}{7}y^1(-k^5)^7 + \binom{8}{8}y^0(-k^5)^8$$

$$\downarrow$$

$$56y^3(-k^{25})$$

$$\downarrow$$

$$-56y^3k^{25}$$

$$\rightarrow \boxed{-56}$$

• ⑬ Find the coefficient of the x^6 term in the expansion of $(x-7)^9$.

$$(x-7)^9 = (x+(-7))^9$$

$$\binom{9}{0}x^9(-7)^0 + \binom{9}{1}x^8(-7)^1 + \binom{9}{2}x^7(-7)^2 + \binom{9}{3}x^6(-7)^3 + \dots + \binom{9}{9}x^0(-7)^9$$

$$\downarrow$$

$$84x^6(-343)$$

$$\downarrow$$

$$-28,812x^6$$

$$\rightarrow \boxed{-28,812}$$

⑭ Find the coefficient of the b^2 term in the expansion of $(b^2+m)^4$.

$$\binom{4}{0}(b^2)^4m^0 + \dots + \binom{4}{3}(b^2)^1m^3 + \binom{4}{4}(b^2)^0m^4$$

$$\downarrow$$

$$4b^2m^3$$

$$\downarrow$$

$$\boxed{4}$$

• ⑮ Find the second term in the expansion of $(x-y)^6$.

$$(x-y)^6 = (x+(-y))^6$$

$$\binom{6}{0}x^6(-y)^0 + \binom{6}{1}x^5(-y)^1 + \dots + \binom{6}{6}x^0(-y)^6$$

$$\downarrow$$

$$6x^5(-y)$$

$$\downarrow$$

$$\boxed{-6x^5y}$$

⑯ Find the fifth term in the expansion of $(4k^7+L)^9$.

$$\binom{9}{0}(4k^7)^9L^0 + \binom{9}{1}(4k^7)^8L^1 + \binom{9}{2}(4k^7)^7L^2 + \binom{9}{3}(4k^7)^6L^3 + \binom{9}{4}(4k^7)^5L^4 + \dots + \binom{9}{9}(4k^7)^0L^9$$

$$\downarrow$$

$$126(1024)k^{35}L^4$$

$$\downarrow$$

$$\boxed{129,024k^{35}L^4}$$

- ⑪ Find the third term in the expansion of $(x-2y^4)^5$.

$$(x-2y^4)^5 = (x+(-2y^4))^5$$

$$\binom{5}{0}x^5(-2y^4)^0 + \binom{5}{1}x^4(-2y^4)^1 + \binom{5}{2}x^3(-2y^4)^2 + \dots + \binom{5}{5}x^0(-2y^4)^5$$

$$\downarrow$$
$$10x^3(4y^8)$$

$$\downarrow$$
$$\boxed{40x^3y^8}$$

- ⑫ Find the last term in the expansion of $(2a+b)^8$.

$$\binom{8}{8}(2a)^0b^8$$

$$\downarrow$$
$$(1)(1)b^8$$

$$\downarrow$$
$$\boxed{b^8}$$